

AetherMind AI — RAG Masterclass

Complete Enterprise Architecture, Concepts & Implementation Manual

1. Introduction to Retrieval-Augmented Generation

Retrieval-Augmented Generation (**RAG**) is an architecture that couples large language models with external information retrieval systems. Standard LLMs are constrained by training cutoffs and lack access to private enterprise data. RAG addresses this by indexing documents into vector space and fetching relevant excerpts dynamically during chat queries.

■ **Key Takeaway:** RAG provides 100% grounded answers with exact source citations without needing to re-train or fine-tune LLM model weights.

2. Core Concepts & Fundamentals

2.1 Document Extraction & Normalization

Raw PDF, DOCX, TXT, and Markdown files are ingested, stripped of non-printable control characters, and normalized. Double-newlines separate paragraphs while keeping page numbers intact for citation tracking.

2.2 Text Chunking & Overlap Math

Documents are split into overlapping word windows (e.g., 300 words with 50 words overlap). Overlap ensures that thoughts spanning across chunk boundaries are not lost in vector space.

Overlap Math Formula: $\text{Step Size} = \text{Chunk Size} - \text{Overlap Size}$

Example: $300 \text{ size} - 50 \text{ overlap} = 250 \text{ word step between window starts}$.

2.3 Vector Embeddings & Asymmetric Search

Embeddings map text into 768-dimensional float vectors. Asymmetric task types (**RETRIEVAL_DOCUMENT** vs **RETRIEVAL_QUERY**) optimize the vector projection for storage vs query matching.

2.4 Similarity Search & HNSW Indexing

ChromaDB computes L2 Euclidean distances between vectors. We convert L2 distance into a normalized 0–1 similarity score: **Similarity** = $1.0 / (1.0 + \text{L2_Distance})$.

2.5 Query Rewriting & Multi-Turn Memory

Context-dependent questions (e.g., 'When was it launched?') are automatically rewritten into standalone search queries ('When was project AetherMind launched?') before embedding lookup.

2.6 Grounding & Safety Heuristics

Strict prompt guardrails force Gemini to answer strictly from retrieved context and append [Source N] tags. Grounding safety checks inspect generated responses for citation compliance.

3. File-by-File Technical Deep Dive

File Path	Core Purpose	Key Technologies
app/config.py	Centralized settings & startup checks	Pydantic BaseSettings, dotenv
app/exceptions.py	Custom application exception hierarchy	RAGBaseError base class
app/logging_config.py	Structured logging & timing context manager	Rich console, duration logger
app/models.py	Domain data models & API schemas	Pydantic V2, Enums
app/ingestion.py	Parsing, SHA-256 dedup, window chunking	pypdf, python-docx, hashlib
app/embeddings.py	Parallel 1-to-1 chunk embedding pipeline	google-genai, ThreadPoolExecutor
app/vectorstore.py	ChromaDB vector store & KB analytics	ChromaDB, PersistentClient
app/chat_memory.py	SQLite chat history, title & export	SQLite3, Markdown export
app/rag.py	Query rewriting, streaming RAG, grounding	asyncio.Queue, Gemini 2.5 Flash
app/main.py	FastAPI REST endpoints & NDJSON stream	FastAPI, SlowAPI, CORS
static/index.html	Glassmorphism UI & Knowledge Explorer	Vanilla JS, Prism.js, Marked.js

4. Key Implementation Highlights

Parallel Embedding Pipeline (`app/embeddings.py`)

To prevent vector count mismatch errors when indexing large PDF files, `embed_documents()` executes individual chunk embeddings concurrently using `ThreadPoolExecutor`. This guarantees exact 1-to-1 matching between input chunks and returned 768-dimensional vector embeddings while offering high throughput.

Async Token Streaming (`app/rag.py` & `app/main.py`)

Token-by-token streaming is achieved by running Gemini's blocking iterator in a producer thread that feeds tokens into an `asyncio.Queue`. The FastAPI endpoint consumes from the queue asynchronously, streaming NDJSON lines to the browser without blocking the main event loop.

■ **Key Takeaway:** *AetherMind AI v2.0.0 represents a complete production-ready RAG platform with unit test coverage across all layers.*